MIDNIGHT HACKATHON

BidShield

Sealed-bid procurement you can prove.

A zero-knowledge tender workflow on Midnight. Suppliers seal prices and documents before close. After the window, anyone can verify that a reveal matches the original commitment — without trusting a database not to leak, swap, or forget.

Watch the demo

youtu.be/dUg0MWxeJH8

Open the repo

github.com/bammyoly/bidshield

Stack

Midnight · Compact · Lace · React

Built for buyers who must stay fair, and suppliers who cannot afford to show their hand.

THE PROBLEM

Public procurement still runs on trust

Sealed envelopes were invented so rivals could not copy a price. Software quietly undid that. Most e-procurement portals store the bid in a database an insider can read, a supplier can later “correct,” or an auditor cannot reconstruct years later.

Bid leaking

Insiders see submitted prices and pass the floor to a favored vendor. The “sealed” bid was never sealed from the operator.

Late alterations

After rivals surface, a supplier claims the original figure was lower. Without a commitment, there is nothing to contradict them.

Document swapping

The winning proposal in the file store is not the one that was submitted. The award is justified after the fact.

No durable trail

Years later, nobody can prove who bid what, when, or whether the winner even submitted before close.

OECD estimates put public procurement fraud and corruption in the region of $2 trillion a year. BidShield does not claim to end that. It removes the easiest cheat: seeing, changing, or denying a bid after it was made.

WHO THIS IS FOR

Three parties, three different fears

Suppliers

A unit price is a trade secret. Showing it early invites undercutting, retaliation, and leaked margins. They need the bid to stay private until close — then they need the buyer to be able to prove it was the bid they actually sent.

Buyers & organizers

They must evaluate compliance, not just the lowest number. They need a record that they did not peek, did not accept a late change, and did not award a packet that was never committed.

Reviewers & the public

Watchdogs, auditors, and losing vendors need a trail that does not depend on the buyer’s database still existing, or on anyone’s word.

THE SOLUTION

A cryptographic commitment layer

One Compact contract instance is one tender. Time windows are sealed at construction. The chain never stores the price at commit time — only a hash the supplier can later open.

1 Commit

The supplier proves they know amount, salt, and document hash. The ledger writes a tender-scoped pseudonym and a persistentHash commitment. Nothing a rival can copy.

2 Reveal

After commitEnd, the same packet is re-hashed. If it matches, amount and document hash become public. If it does not match, the circuit refuses.

3 Award

After revealEnd, the organizer records a winner who already revealed, plus a hash of the award rationale. The decision is attested — not auto-scored.

Best value, not auto-lowest. Compliance documents stay off-chain (only their hash is committed). Buyers still read the files, disqualify bids that fail policy, then lock the winner on Midnight. The chain proves the packet is the original one. It does not pretend to have read the PDF.

PRIVACY BY DESIGN

What stays hidden, and what must become public

commitment = persistentHash(tag · tenderId · bidder · amount · salt · documentHash)

Before the deadline

Public: tender policy hash, time windows, a tender-scoped pseudonym, and the commitment.

Private: amount, salt, supporting file, and the bidder secret. Even the organizer cannot read prices. The secret is derived in the browser from the Lace account and tender id — never written as a wallet address.

After a valid reveal

Public: the disclosed amount and document hash, checked against the stored commitment.

Still off-chain: the file bytes, legal identity, and contact details. Revealed prices are public in this MVP — that is a product choice, not a leak. Phases are time-locked: no late bids, no edits after seeing rivals.

A matching reveal proves “this is the packet I sealed.” It does not prove the price is truthful, the document is authentic, or the buyer chose fairly. BidShield labels those as attestations, not circuit facts.

WHY MIDNIGHT

The only place this design is native

Private inputs, public consequences

Compact circuits take amount, salt, and hashes as witnesses. The proof is public. The values never have to sit in a server log for the check to work.

Time as a gate, not a timestamp field

blockTimeGte / blockTimeLt lock commit, reveal, and award. Nobody can open early or sneak a bid in after close by editing a row.

Tender-scoped identity

The bidder key is hash(tag, tenderId, secret). Lace binds a stable local secret without publishing mn_addr_… as the bidder.

Selective disclosure

disclose() is used only on the commitment and, later, on the revealed amount and document hash. That is the product: hide, then prove, then show.

ARCHITECTURE

Chain for truth. Client for secrets. App for the work.

Midnight · Compact

Source of truth. Constructor seals organizer key, tender id, policy hash, and unix-second windows. Circuits: commitBid, revealBid, recordAward. Live Preview instance; artifacts copy into the frontend.

Lace wallet

DApp Connector v4: connect('preview'). Signs and proves in the wallet against a local proof server. BidShield never treats the unshielded address as the on-chain bidder id.

React console

Glass audit UI: live windows, indexer action, commit/reveal/award flows, local packet history. Documents are hashed in the browser; the file itself stays in encrypted off-chain storage.

If an application database is wiped, bid integrity is unaffected: the commitments live on Midnight. The app is a lens and a proving client, not the ledger.

DEMO

End-to-end in four beats

01 Connect & commit

Supplier links Lace on Preview, enters amount, hashes a document, keeps the salt. commitBid writes only the commitment.

02 Window flips

When commitEnd passes, new commitments are refused. Reveal opens from the clock — not from an admin toggle that could be gamed.

03 Reveal

The same packet is loaded. The circuit checks the hash. On success the amount and document hash go public.

04 Award

Organizer secret must match the constructor. Winner must already have revealed. Rationale hash is recorded as an attestation.

Watch it happen

Video walkthrough → youtube.com/watch?v=dUg0MWxeJH8

Source and deploy notes → github.com/bammyoly/bidshield

WHAT MAKES IT DIFFERENT

Fair enough to audit. Flexible enough to award.

Not an auto-lowest black box

Procurement is rarely “min(price).” Buyers still read compliance packs off-chain. BidShield forces the price they award to be a price that was sealed on time — then lets humans do the rest.

Organizer is a proof, not a login

recordAward witnesses localOrganizerSecret against the key hashed at construction. Matching a deployer address in the UI is only a convenience gate. The circuit is the authority.

Honest about what ZK cannot do

The product copy refuses to claim fairness, authenticity, or legal validity. Reviewers see contract-verified facts vs organizer attestations vs off-chain files. That honesty is the audit feature.

Public, timestamped trail

Commitments, reveals, and awards sit on Preview for anyone with the address. Explorers and the indexer are first-class, not a private admin screen.

HONEST STATUS

Working today — and what we will not pretend

Shipped in this hackathon

- Compact contract: commitBid, revealBid, recordAward
- Deployed on Midnight Preview with constructor windows
- React + TypeScript console with glass audit UI
- Lace DApp Connector (v4 connect, legacy enable fallback)
- Tender-scoped local bidder secret from the wallet
- Time-gated phases and indexer-backed tender view
- Local sealed-packet history in the browser

Circuits are real Compact. Submitting them still needs Lace plus a local proof server — that is Midnight today, not a BidShield shortcut.

Next, in order

1. End-to-end circuit submit from the console (prove + Lace sign).
2. Encrypted off-chain document store (hash on-chain, bytes elsewhere).
3. Indexer history of every commit/reveal, not only latest action.
4. Optional supplier registry so one party cannot farm many secrets.
5. Multi-tender factory — still one instance per tender.

Out of scope for MVP: escrow, scoring models, legal workflow, dispute courts.

USE CASES

Anywhere a price must stay dark until close

Government & enterprise

Infrastructure, health, and IT tenders where leaking a unit price is leaking a margin — and leaking an award process is a headline.

DAO treasuries & grants

Members should not see each other’s asks before the round closes. After close, the DAO can show the matching reveals.

Web3 subcontracting

Security audits, protocol work, market-making. Vendors bid privately; the award hash is a public receipt.

Internal reverse auctions

A company can run a sealed reverse auction among approved vendors without teaching them each other’s floors.

THANK YOU

BidShield

Bids that cannot be leaked, altered, or denied.

Hide the price until close. Keep a verifiable trail after. Let humans still choose. Let the chain only claim what it can prove.

Repository

github.com/bammyoly/bidshield

Video demo

youtu.be/dUg0MWxeJH8

Midnight Preview · Compact 0.31 · Lace · React